Exercise: Applying the Pioneer Detection Method to European Inflation Dynamics

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Based on the Pioneer Detection Method (PDM) introduced in:

Vansteenberghe, E. (2025), "Insurance Supervision under Climate Change: A Pioneer Detection Method," The Geneva Papers on Risk and Insurance – Issues and Practice.

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Context

The Pioneer Detection Method (PDM) was developed to identify experts whose estimates deviate early from the consensus but toward whom other experts subsequently converge. Originally designed for insurance supervision under climate change, where the true tail parameter of the loss distribution is unobservable, the method applies naturally to any setting where multiple non-cooperating agents produce time series that may exhibit directional convergence after a structural break.

In this exercise, you will apply the PDM and its alternatives to a panel of European inflation time series. Each country's inflation rate is treated as an "expert estimate" of the underlying inflation trend. Countries do not coordinate their inflation outcomes—each responds to its own supply and demand shocks, energy mix, and policy environment. Yet common shocks (energy prices, monetary policy, supply chains) induce convergence patterns that the PDM can detect.

Repository: `https://github.com/skimeur/pioneer-detection-method`
Starter code: `ecb_hicp_panel_var_granger.py` (data fetching and Granger/VAR analysis) and `pdm.py` (all PDM methods).

Data

The inflation panel is built from two sources:

- **ECB Data Portal** (SDMX 2.1): monthly HICP inflation (year-on-year, %) for 11 eurozone countries: DE, FR, IT, ES, NL, BE, AT, PT, IE, FI, GR.
- **State Statistics Service of Ukraine** (SSSU, SDMX v3): monthly CPI (previous month = 100), converted to year-on-year inflation.

The panel covers 2000–2025 at monthly frequency. The script `ecb_hicp_panel_var_granger.py` fetches and merges these series into a single DataFrame `infl_panel` with columns = country codes and rows = months.

Available methods in `pdm.py`

1 Part A: Who pioneered European inflation dynamics?

In this part, treat the full panel of 12 series (11 EU countries + Ukraine) as the set of "experts."

#	Method	Function
1	PDM with angles (preferred)	`compute_pioneer_weights_angles`
2	PDM with distances	`compute_pioneer_weights_distance`
3	Granger Causality	`compute_granger_weights`
4	Lagged Correlation	`compute_lagged_correlation_weights`
5	Multivariate Linear Regressions	`compute_multivariate_regression_weights`
6	Transfer Entropy	`compute_transfer_entropy_weights`
7	Linear Opinion Pooling	`compute_linear_pooling_weights`
8	Median Pooling	`compute_median_pooling`

All weight functions take a `(T × N)` DataFrame and return weights of the same shape (except Median Pooling, which returns a Series). Use `pooled_forecast(forecasts, weights)` to compute the weighted combination.

Question A.1 — PDM weights over time

(a) Load the inflation panel using `ecb_hicp_panel_var_granger.py`. Restrict to the complete-case sample (`.dropna()`).

(b) Apply `compute_pioneer_weights_angles` to the panel. This returns a `(T × 12)` DataFrame of time-varying pioneer weights.

(c) Plot the pioneer weight of each country over time (one line per country, or a heatmap). Which countries receive non-zero pioneer weight, and when?

(d) Discuss: during periods of low and stable inflation (e.g. 2000–2007), do you observe strong pioneers? What does this mean in the context of the PDM theory?

Question A.2 — Average pioneer weights by subperiod

Compute the average PDM (angles) weight for each country over the following subperiods:

Period	Dates	Economic context
I	2002–2007	Great Moderation, low inflation dispersion
II	2008–2012	Global Financial Crisis, eurozone sovereign debt crisis
III	2013–2019	Low inflation / deflation fears, ECB QE
IV	2020–2021	COVID-19 pandemic, supply chain disruptions
V	2022–2023	Energy crisis, post-invasion inflation surge
VI	2024–2025	Disinflation phase

(a) Present the results in a table: rows = countries, columns = subperiods, values = average pioneer weight.

(b) Rank countries in each subperiod. Does the ranking change over time?

(c) Provide an economic interpretation: which structural characteristics (energy mix, trade openness, financial sector size, geographic position) might explain early exposure to inflation shocks?

2 Part B: Predicting Ukraine's inflation trajectory

In this part, treat Ukraine as the "true parameter to predict" and the 11 EU countries as the "experts."

Motivation

Let's imagine you are a supervisor monitoring Ukrainian inflation not observing the structural forces driving it (war, energy markets, agricultural disruption, monetary policy) but can observe EU countries' inflation outcomes. Some EU countries may exhibit inflation dynamics that *precede* similar movements in Ukraine—for instance, because energy price shocks hit EU markets before propagating to Ukraine through trade channels, or because EU monetary policy signals foreshadow global inflation trends.

The goal is to identify which EU countries are "pioneers" for Ukrainian inflation, i.e., which countries' inflation moves first in a direction that Ukraine's inflation subsequently follows.

Question B.1 — Rolling pioneer detection

(a) Apply the PDM (angles) on a rolling window. At each window position, record which EU country has the highest pioneer weight relative to Ukraine.

(b) Plot the "dominant pioneer" over time. Does the identity of the pioneer change across subperiods?

Question B.2 — Forecasting evaluation

(a) For each method, compute the pooled estimate of Ukraine's inflation using only EU countries as inputs (i.e., use the method's weights applied to EU countries' inflation to produce a "forecast" of where Ukraine's inflation is heading).

Formally, the supervisor's estimate at time t is:

$$\hat{\pi}^t_{\text{UA}} = \sum_{i \in \text{EU}} w^t_i \cdot \pi^t_i$$

where π^t_i is country i's inflation and w^t_i are the pioneer weights.

(b) Compare each method's pooled estimate against Ukraine's actual inflation using Root Mean Square Error:

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{t=1}^{T} \left(\hat{\pi}^t_{\text{UA}} - \pi^t_{\text{UA}}\right)^2}$$

(c) Report RMSE for each method and each subperiod. Which method produces the most accurate "forecast" of Ukraine's inflation?

(d) **Important caveat:** The PDM was not designed as a forecasting tool—it was designed to identify pioneers and accelerate collective learning. A low RMSE here does not mean the method is a good forecaster; it means that the weighted combination of EU countries' inflation happens to track Ukraine's inflation well. Discuss the limits of this interpretation.

Code template

The following code connects the data from `ecb_hicp_panel_var_granger.py` to the methods in `pdm.py`:

```
# After running ecb_hicp_panel_var_granger.py, you have infl_panel
# with columns: DE, FR, IT, ES, NL, BE, AT, PT, IE, FI, GR, UA

from pdm import (
    compute_pioneer_weights_angles,
```

```python
    compute_pioneer_weights_distance,
    compute_granger_weights,
    compute_lagged_correlation_weights,
    compute_multivariate_regression_weights,
    compute_transfer_entropy_weights,
    compute_linear_pooling_weights,
    compute_median_pooling,
    pooled_forecast,
)

# --- Part A: full panel, all countries as "experts" ---
panel = infl_panel.dropna()
w_angles = compute_pioneer_weights_angles(panel)

# Average weights by subperiod
periods = {
    "I (2002-07)": ("2002-01", "2007-12"),
    "II (2008-12)": ("2008-01", "2012-12"),
    "III (2013-19)":("2013-01", "2019-12"),
    "IV (2020-21)": ("2020-01", "2021-12"),
    "V (2022-23)": ("2022-01", "2023-12"),
    "VI (2024-25)": ("2024-01", "2025-12"),
}
for name, (s, e) in periods.items():
    sub = w_angles.loc[s:e]
    print(f"\n{name}")
    print(sub.mean().sort_values(ascending=False))

# --- Part B: EU countries as experts, Ukraine as target ---
eu_cols = [c for c in panel.columns if c != "UA"]
eu_panel = panel[eu_cols]
w_eu = compute_pioneer_weights_angles(eu_panel)
forecast_ua = pooled_forecast(eu_panel, w_eu)

# RMSE vs actual Ukraine inflation
import numpy as np
actual_ua = panel["UA"]
rmse = np.sqrt(((forecast_ua - actual_ua) ** 2).mean())
print(f"RMSE (PDM angles): {rmse:.4f}")
```

Deliverables

1. **A dedicated Python script** named `exercise_pdm_inflation.py` containing all your code. This script should:
   - import from `pdm.py` and reuse data-fetching logic from `ecb_hicp_panel_var_granger.py`;
   - be self-contained: running `python exercise_pdm_inflation.py` should produce all tables and figures;
   - be clearly structured with sections matching Parts A and B.
2. **Tables**: average pioneer weights by country and subperiod (Question A.2), RMSE by method (Question B.2).
3. **Figures**: pioneer weights over time (Question A.1), dominant pioneer over time (Question B.1).

4. **A short written discussion** of your results: economic interpretation of the pioneer rankings (Part A) and limits of the forecasting analogy (Part B, Question B.2d).

Submission

When your code is complete and produces correct results:

1. Fork the original repository at `https://github.com/skimeur/pioneer-detection-method`.
2. Add your `exercise_pdm_inflation.py` script (and any figures or tables you wish to include) to your fork.
3. Open a **Pull Request** to the original repository with:
   - a clear title, e.g. `[Exercise] PDM inflation panel - Your Name`;
   - a short description of your approach and main findings in the PR body.

This allows the course instructor to review your work directly on GitHub and provides you with experience contributing to an open-source project.